

LFX Linux Kernel Mentorship — Spring 2026

Final Report — Sanjay Chitroda

Mentorship: 1 March 2026 – 31 August 2026

Mentors: Shuah Khan & Brigham Campbell

Primary Subsystem: Industrial I/O (IIO)

During the mentorship I focused on upstream Linux kernel development in the IIO subsystem. My work primarily involved driver modernization, resource management improvements, lifecycle fixes, error-handling enhancements and MAINTAINERS updates.

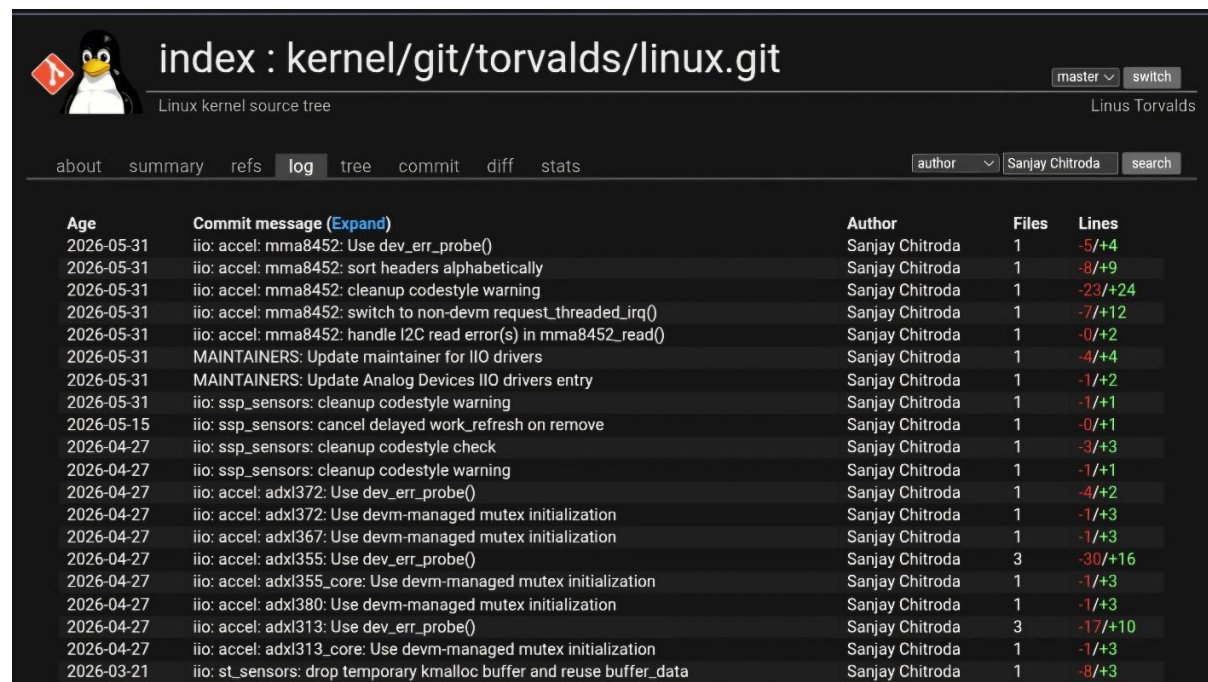
The LFX graduation requirement is a minimum of five accepted upstream patches. During the mentorship, **20 commits were accepted into mainline Linux** and **27 additional commits progressed through linux-next**, giving a total of **47 contributions** represented in this report.

The mentorship provided practical experience with mailing-list development, patch review, maintainer feedback, linux-next tracking and upstream development workflows. My work was not based on Syzbot issues; instead it focused on preventive maintenance and modernization to reduce complexity and potential future failure paths.

The following sections provide author-log snapshots from mainline Linux and linux-next during the mentorship period.

Accepted Mainline Contributions

20 commits during the mentorship period.

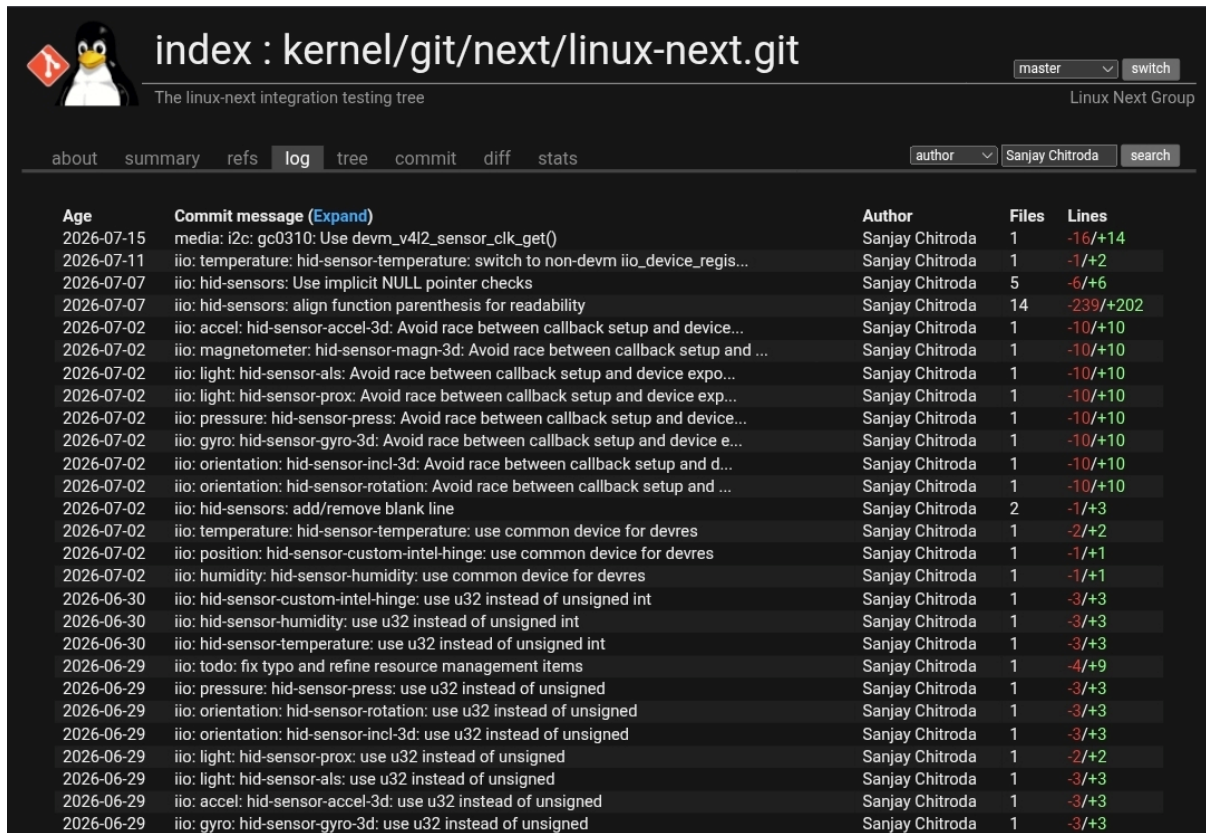


The screenshot shows the 'index : kernel/git/torvalds/linux.git' page for the Linux kernel source tree. It displays a list of 20 commits by Sanjay Chitroda, sorted by age. The table includes columns for Age, Commit message, Author, Files, and Lines. The commits are all from the 'iio' subsystem and include updates to MAINTAINERS, driver modernizations, and resource management improvements.

Age	Commit message (Expand)	Author	Files	Lines
2026-05-31	iio: accel: mma8452: Use dev_err_probe()	Sanjay Chitroda	1	-5/+4
2026-05-31	iio: accel: mma8452: sort headers alphabetically	Sanjay Chitroda	1	-8/+9
2026-05-31	iio: accel: mma8452: cleanup codestyle warning	Sanjay Chitroda	1	-23/+24
2026-05-31	iio: accel: mma8452: switch to non-devm request_threaded_irq()	Sanjay Chitroda	1	-7/+12
2026-05-31	iio: accel: mma8452: handle I2C read error(s) in mma8452_read()	Sanjay Chitroda	1	-0/+2
2026-05-31	MAINTAINERS: Update maintainer for IIO drivers	Sanjay Chitroda	1	-4/+4
2026-05-31	MAINTAINERS: Update Analog Devices IIO drivers entry	Sanjay Chitroda	1	-1/+2
2026-05-31	iio: ssp_sensors: cleanup codestyle warning	Sanjay Chitroda	1	-1/+1
2026-05-15	iio: ssp_sensors: cancel delayed work_refresh on remove	Sanjay Chitroda	1	-0/+1
2026-04-27	iio: ssp_sensors: cleanup codestyle check	Sanjay Chitroda	1	-3/+3
2026-04-27	iio: ssp_sensors: cleanup codestyle warning	Sanjay Chitroda	1	-1/+1
2026-04-27	iio: accel: adxl372: Use dev_err_probe()	Sanjay Chitroda	1	-4/+2
2026-04-27	iio: accel: adxl372: Use devm-managed mutex initialization	Sanjay Chitroda	1	-1/+3
2026-04-27	iio: accel: adxl367: Use devm-managed mutex initialization	Sanjay Chitroda	1	-1/+3
2026-04-27	iio: accel: adxl355: Use dev_err_probe()	Sanjay Chitroda	3	-30/+16
2026-04-27	iio: accel: adxl355_core: Use devm-managed mutex initialization	Sanjay Chitroda	1	-1/+3
2026-04-27	iio: accel: adxl380: Use devm-managed mutex initialization	Sanjay Chitroda	1	-1/+3
2026-04-27	iio: accel: adxl313: Use dev_err_probe()	Sanjay Chitroda	3	-17/+10
2026-04-27	iio: accel: adxl313_core: Use devm-managed mutex initialization	Sanjay Chitroda	1	-1/+3
2026-03-21	iio: st_sensors: drop temporary kmalloc buffer and reuse buffer_data	Sanjay Chitroda	1	-8/+3

Additional linux-next Contributions

27 additional contributions progressed through linux-next.



The screenshot shows the GitHub interface for the `linux-next` repository. The header includes the Linux logo, the repository name `index : kernel/git/next/linux-next.git`, and a dropdown menu set to `master`. Below the header, there are tabs for `about`, `summary`, `refs`, `log` (selected), `tree`, `commit`, `diff`, and `stats`. A search bar is also present. The main content area displays a list of commits, with columns for `Age`, `Commit message (Expand)`, `Author`, `Files`, and `Lines`. All commits are by Sanjay Chitroda and relate to the `iio` subsystem, specifically focusing on sensor drivers and race conditions.

Age	Commit message (Expand)	Author	Files	Lines
2026-07-15	media: i2c: gc0310: Use devm_v4l2_sensor_clk_get()	Sanjay Chitroda	1	-16/+14
2026-07-11	iio: temperature: hid-sensor-temperature: switch to non-devm iio_device_regis...	Sanjay Chitroda	1	-1/+2
2026-07-07	iio: hid-sensors: Use implicit NULL pointer checks	Sanjay Chitroda	5	-6/+6
2026-07-07	iio: hid-sensors: align function parenthesis for readability	Sanjay Chitroda	14	-239/+202
2026-07-02	iio: accel: hid-sensor-accel-3d: Avoid race between callback setup and device...	Sanjay Chitroda	1	-10/+10
2026-07-02	iio: magnetometer: hid-sensor-magn-3d: Avoid race between callback setup and ...	Sanjay Chitroda	1	-10/+10
2026-07-02	iio: light: hid-sensor-als: Avoid race between callback setup and device expo...	Sanjay Chitroda	1	-10/+10
2026-07-02	iio: light: hid-sensor-prox: Avoid race between callback setup and device exp...	Sanjay Chitroda	1	-10/+10
2026-07-02	iio: pressure: hid-sensor-press: Avoid race between callback setup and device...	Sanjay Chitroda	1	-10/+10
2026-07-02	iio: gyro: hid-sensor-gyro-3d: Avoid race between callback setup and device e...	Sanjay Chitroda	1	-10/+10
2026-07-02	iio: orientation: hid-sensor-incl-3d: Avoid race between callback setup and d...	Sanjay Chitroda	1	-10/+10
2026-07-02	iio: orientation: hid-sensor-rotation: Avoid race between callback setup and ...	Sanjay Chitroda	1	-10/+10
2026-07-02	iio: hid-sensors: add/remove blank line	Sanjay Chitroda	2	-1/+3
2026-07-02	iio: temperature: hid-sensor-temperature: use common device for devres	Sanjay Chitroda	1	-2/+2
2026-07-02	iio: position: hid-sensor-custom-intel-hinge: use common device for devres	Sanjay Chitroda	1	-1/+1
2026-07-02	iio: humidity: hid-sensor-humidity: use common device for devres	Sanjay Chitroda	1	-1/+1
2026-06-30	iio: hid-sensor-custom-intel-hinge: use u32 instead of unsigned int	Sanjay Chitroda	1	-3/+3
2026-06-30	iio: hid-sensor-humidity: use u32 instead of unsigned int	Sanjay Chitroda	1	-3/+3
2026-06-30	iio: hid-sensor-temperature: use u32 instead of unsigned int	Sanjay Chitroda	1	-3/+3
2026-06-29	iio: todo: fix typo and refine resource management items	Sanjay Chitroda	1	-4/+9
2026-06-29	iio: pressure: hid-sensor-press: use u32 instead of unsigned	Sanjay Chitroda	1	-3/+3
2026-06-29	iio: orientation: hid-sensor-rotation: use u32 instead of unsigned	Sanjay Chitroda	1	-3/+3
2026-06-29	iio: orientation: hid-sensor-incl-3d: use u32 instead of unsigned	Sanjay Chitroda	1	-3/+3
2026-06-29	iio: light: hid-sensor-prox: use u32 instead of unsigned	Sanjay Chitroda	1	-2/+2
2026-06-29	iio: light: hid-sensor-als: use u32 instead of unsigned	Sanjay Chitroda	1	-3/+3
2026-06-29	iio: accel: hid-sensor-accel-3d: use u32 instead of unsigned	Sanjay Chitroda	1	-3/+3
2026-06-29	iio: gyro: hid-sensor-gyro-3d: use u32 instead of unsigned	Sanjay Chitroda	1	-3/+3

References

- Mainline: <https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git/log/?qt=author&q=sanjay+chitroda>
- linux-next: <https://git.kernel.org/pub/scm/linux/kernel/git/next/linux-next.git/log/?qt=author&q=sanjay+chitroda>
- Mentorship repository: <https://github.com/sanjay-embedded/LFX-Linux-kernel-Spring-2026>
- Blog: <https://sanjay-embedded.github.io/LFX-Linux-kernel-Spring-2026/>